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Note	

EEEN10131 Digital System Design I

Laboratory 2 – Sequential Logic

Getting Started

This laboratory will be run in the same way as the first one, in that you will be using the NI myDAQ in your own time with either your own personal computer or those in university's clusters. If you used your own computer the first time you will already have the software.

In this laboratory you will be using the NI ELVISmx Digital Writer and Digital Reader. If you need more help with using these virtual instruments, refer back to the instructions in the first laboratory of DSD1.

Essential Lab Preparation

This laboratory involves the design of sequencer similar to the one illustrated in the example given below.

Follow the example and make sure you understand the following.

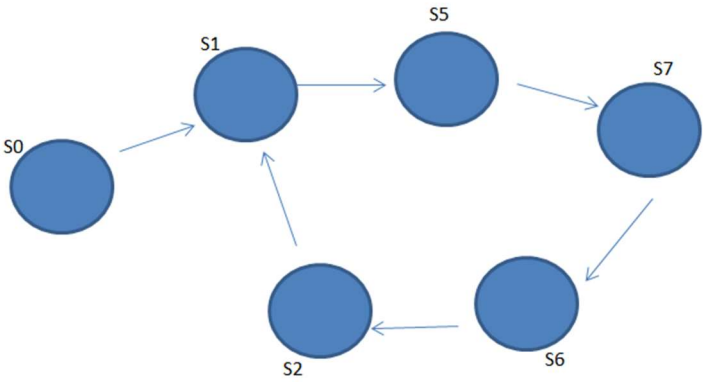
- Drawing the state diagram according to the specification
- Completing the state transition table
- Drawing the Karnaugh Maps and deriving a minimised next state logic for the sequencer
- Drawing the circuit diagram of the derived logic equation including all the required information.

The script was initiated by Amanda Weerasinghe

Specification:
Starting from state S0, the sequencer should cycle through the states S1, S5, S7, S6, S2, S1, S5, S7,…… (See State Diagram below).

The coding for each state should be 3-bit binary equivalent.

State Diagram:



State Transition Table:
Label the bits of the present state registers as Q_3, Q_2, $Q_1$and the next state bits as D_3, D_2 and D_1.

Present State			Next State		
Q_3	Q_2	Q_1	D_3	D_2	D_1
0	0	0	0	0	1
0	0	1	1	0	1
0	1	0	0	0	1
0	1	1	X	X	X
1	0	0	X	X	X
1	0	1	1	1	1
1	1	0	0	1	0
1	1	1	1	1	0

Note: The states “011” and “100” are not in the sequence. Therefore, when the present states are “011” and “100”, we “do not care” what the corresponding next state bits are.

However, it is important to use the “don’t care” terms in the Karnaugh maps to find the optimum solution for the minimised next state logic expression.

Combinatorial logic:

Q_3 \ Q_2 Q_1	00	01	11	10
0	0	1	X	0
1	X	1	1	0

K-map for D_3 ; $D_3 = Q_1$

Q_3 \ $Q_2 Q_1$	00	01	11	10
0	0	0	X	0
1	X	1	1	1

K-map for D_2 ; $D_2 = Q_3$

Q_3 \ $Q_2 Q_1$	00	01	11	10
0	1	1	X	1
1	X	1	0	0

K-map for D_1 ; $D_1 = \overline{Q_3} + \overline{Q_2} = \overline{Q_3 \cdot Q_2}$

Circuit Diagram:
The following information must be included to complete a logic diagram for this laboratory.

- Identification of the chips that are used for each gate
- Pin numbers on the integrated circuits corresponding to the inputs and outputs of each gate
- Names of the connections to the NI myDAQ (e.g. DIO 0 etc.)

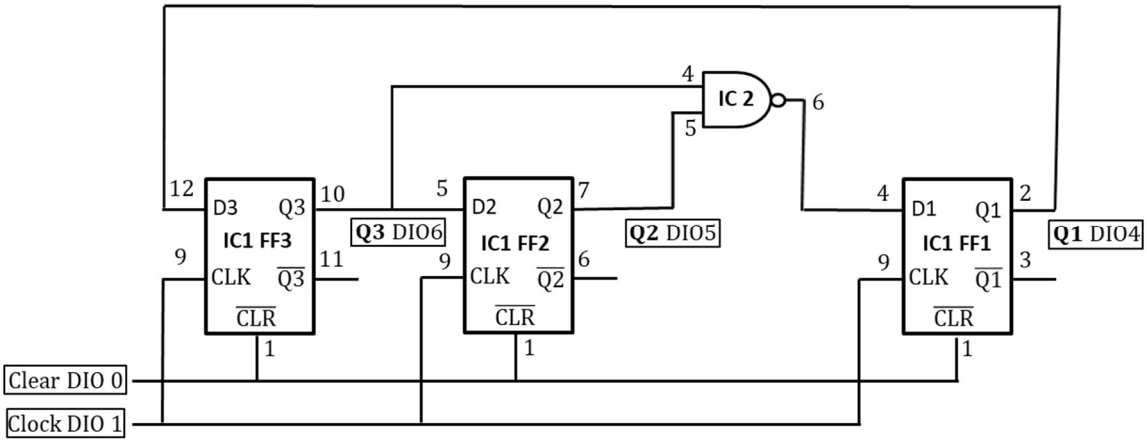


Figure 1: The circuit diagram for the example sequencer

Timing Diagrams:
You will be expected to draw graphs representing the change of logic levels that you observe. The levels will be either “high” or “low”. An example is shown below.



The logic level is indicated on the vertical axis and the time on the horizontal axis. “t0”, “t1”, “t2” and “t3” represent time-steps. In this example, the signal “X” which could be an input or output is initially low at time t0. It goes high at time t1. It stays high until time t3.

Part A: D-type Latch

This part of the lab uses the chip CD4042BE by Texas Instruments. This integrated circuit comprises 4 D-type latches. The “D” pins are the inputs while the “Q” pins are the outputs.

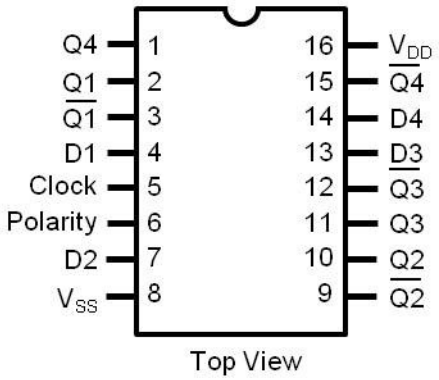


Figure 2: Pin assignment of CD4042BE by Texas Instruments.

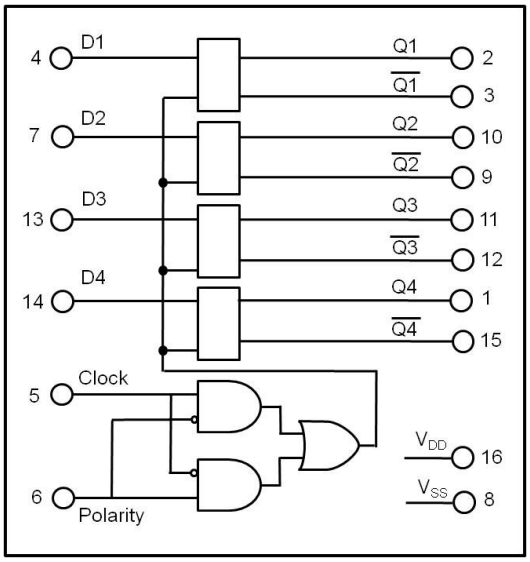


Figure 3: Functional Diagram of CD4042BE by Texas Instruments.

Note: Complete the building of the circuit before powering on the myDAQ breadboard. This is always good practice.

- Place CD4042BE on the breadboard. Use “+” and “-” rails as power rails on the breadboard to connect 5 V and 0 V (Ground) respectively.
- Connect V_{DD}, Pin 16 to 5 V. (Positive supply voltage must be connected to V_{DD} in CMOS technology while it is Vcc for TTL. We are using a CMOS device.)
- Connect Vss, Pin 8 to Ground.
- Note that Vss and V_{DD} pins supply power to the chip.

Important: You should double check that you have connected the correct pins to the power supply before powering on the myDAQ. Otherwise, you could damage the chips.

- Now wire up the input and output pins of the chip using the DIO (digital I/O) lines on the myDAQ. The connections are summarised in the table below.

Function	Pin number on CD4042BE	Which line the pin should be connected to on myDAQ
Inputs to the chip		
Clock	5	DIO 0
D3 (logic input)	13	DIO 1
Polarity	6	5 V power rail
Outputs from the chip		
Q3 (logic output)	11	DIO 7

- Connecting the Polarity pin to 5 V ensures that the latches are active when the clock is high. If the Polarity pin is connected to 0 V the latches will be active low.
- Connect the NI myDAQ to the computer and ensure that it is recognised by the computer (more details on configuring NI myDAQ can be found in the lab script for the 1st lab). Power on the breadboard by switching on the 5 V switch on the breadboard.

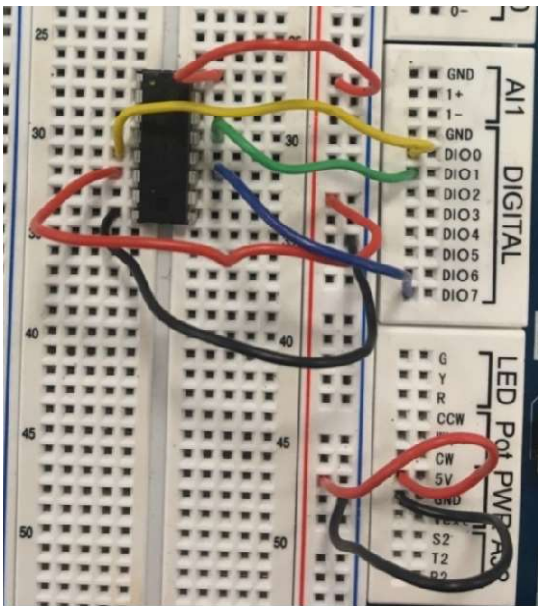


Figure 4: Placing the D-type latch on the breadboard and wiring up the circuit. Note that 5 V and GND are connected to "+" and "-" rails.

Important: Ensure that the black 5 V switch on the breadboard is at the 'ON' position. Otherwise the circuit will not receive power and will not give the correct output.

- Open Digital Reader and Digital Writer from the NI ELVISmx Instrument Launcher.
- Configure digital lines as inputs to the myDAQ (outputs from the circuit you have built) by choosing the required group of lines in the Digital Reader. The output from the D type latch (Q3) is connected to DIO7. Hence, choose 'Lines to Read' as 4-7 in the Digital Reader.
- Configure digital line outputs from the myDAQ (inputs to the circuit you have built) by choosing the required group of lines in the Digital Writer. Inputs to the D type latch are connected to DIO0 and DIO1. Hence, choose 'Lines to Write' as 0-3 in the Digital Writer.



Figure 5: How to configure Digital Reader and Digital Writer. Note the "Lines to Write" and "Lines to Read" drop down menus of the Digital Writer and Digital Reader respectively.

Important: You cannot configure one line as both an input and output simultaneously as this will cause a conflict between Digital Reader and Digital Writer.

- Make DIO 0 and DIO 1 logic low by setting the lines to position "LO" on the Digital Writer. This is done to ensure than the clock and D3 input to the latch are logic low initially.
- Run both Digital Reader and Digital Writer.
- Carefully record the values on Q3 from the **Digital Reader's line 7** as a result of the change made to line 0 (Clock) and line 1 (D3) on the **Digital Writer** at the **chronological order (time steps)** given in the table below. (Note that LO= logic low and HI= logic high)

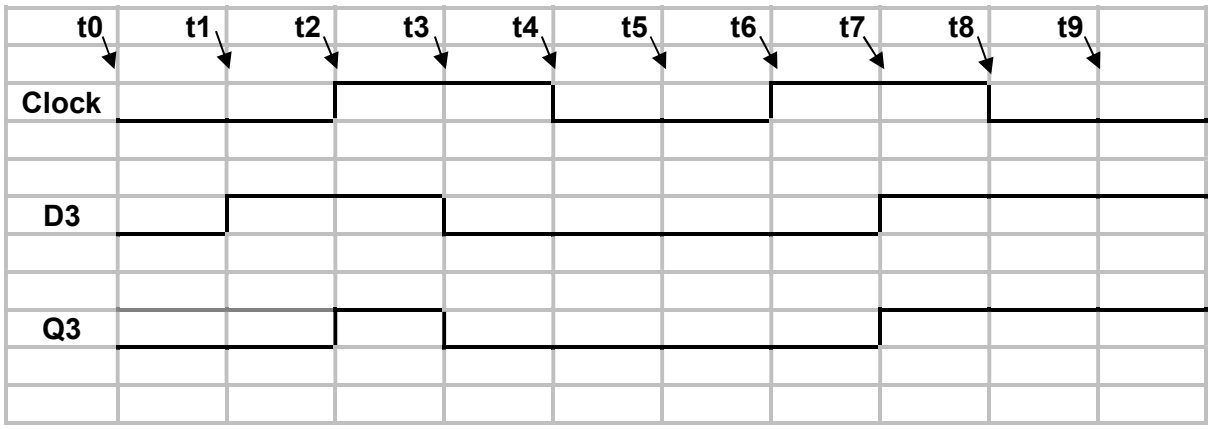
Time step	Inputs (change the values on the Digital Writer as follows)		Output Q3 (read the value from line 7 of Digital Reader)
	D3 (line 1)	Clock (line 0)	
t0	LO	LO	LO
t1	HI	Unchanged from last	LO
t2	Unchanged from last	HI	HI
t3	LO	Unchanged from last	LO
t4	Unchanged from last	LO	LO
t5	Unchanged from last	Unchanged from last	LO
t6	Unchanged from last	HI	LO
t7	HI	Unchanged from last	HI
t8	Unchanged from last	LO	HI
t9	Unchanged from last	Unchanged from last	HI

- Before proceeding with the lab, consider the sequence of transitions that you have just recorded. If these are not what you would expect from a D-type latch then you should double check your circuit and procedure of recording the data.

Task 1

(8 marks)

Using the values you have recorded for the D-type latch, plot the graphs of the inputs (Clock, D3) and output (Q3) logic levels as a function of time.

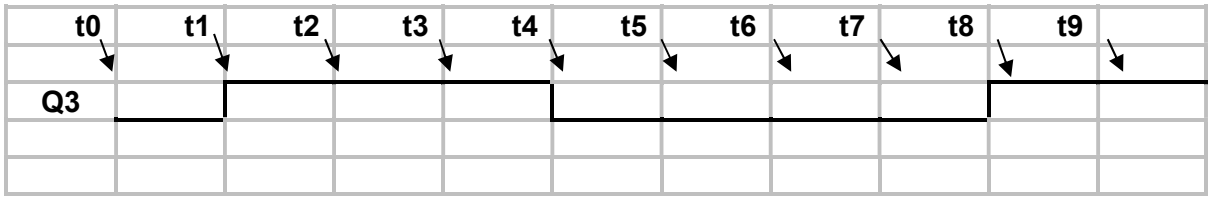


Task 2

(6 marks)

Consider the transitions on Clock and D3 for the D-type latch you have recorded above. The D-type latch we have used is active high meaning that the latch is active when the clock is high. What would be the output (Q3) be if the latch is active when the clock is low (active low). Plot the timing diagram for Q3 if the latch is active low.

You can first predict the output and then connect the Polarity pin to Ground rail and perform the experiment again to verify your answer. Ensure that you switch off the power to the breadboard before rewiring.



Part B: D-type Flip Flop

This part of the lab uses chip CD40175BE which consists of 4 D-type Flip-Flops. The “D” pins are the inputs while the “Q” pins are the outputs.

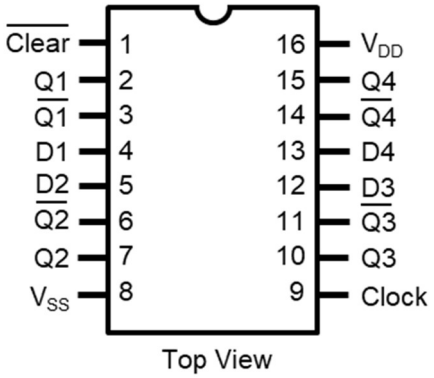


Figure 6: Pin assignment for CD40175BE by Texas Instruments.

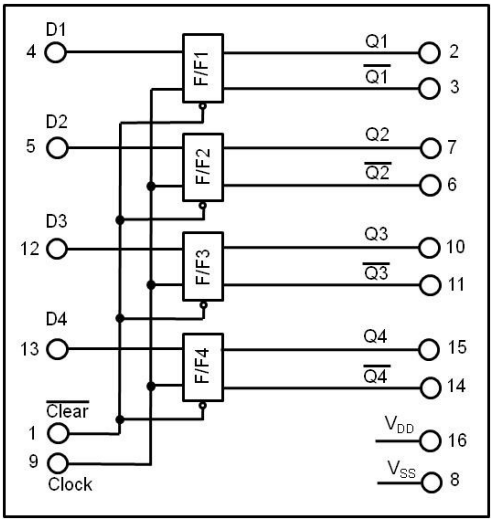


Figure 7: Functional Diagram for CD40175BE by Texas Instruments

Note: Complete the building of the circuit before powering on the myDAQ. This is always good practice.

- Unplug myDAQ and remove all the wires to the Quad D-type latch.
- Place the CD40175BE on the breadboard. Use “+” and “-“ rails on the breadboard as power rails to connect 5 V and 0 V (Ground) respectively.
- Connect V_{DD}- Pin 16 to 5 V. (The positive supply voltage must be connected toV_{DD} in CMOS technology while it is Vcc for TTL)
- Connect the Vss- Pin 8 to Ground.
- Note that the Vss and V_{DD} pins supply power to the chip.

Important: You should double check that you have connected the correct pins to the power supply before powering on the myDAQ. Otherwise, you could damage the chips.

- Now we should wire up the inputs and output pins of the chip. We will be using DIO (digital I/O) lines on the myDAQ. The connections are summarised in the table below.

Function	Pin number on CD40175BE	Which line the pin should be connected to on myDAQ
Inputs to the chip		
Clock (C)	9	DIO 0
D4 (logic input)	13	DIO 1
$\overline{Clear}$	1	5 V power rail
Outputs from the chip		
Q4 (logic output)	15	DIO 7

- Pin 1 should be logic high (5 V) to allow the flip-flop to function. If it is logic low, the Q outputs are reset to logic low (cleared) regardless of the values on Clock and D inputs.
- Connect the NI myDAQ to the computer and ensure that it is recognised by the computer (more details can be found in the lab script for the 1st lab). Power on the breadboard by switching on 5 V switch on the breadboard.

Important: Ensure that the black 5 V switch on the breadboard is at the 'ON' position. Otherwise the circuit will not receive power and will not give the correct output.

- Open Digital Reader and Digital Writer from NI ELVISmx Instrument Launcher.
- Configure digital lines as inputs to the myDAQ (outputs from the circuit you have built) by choosing the required group of lines in the Digital Reader. The output from the D type flip-flop (Q4) is connected to DIO7. Hence, choose 'Lines to Read' as 4-7 in the Digital Reader.
- Configure digital lines as outputs from the myDAQ (inputs to the circuit you have built) by choosing the required group of lines in the Digital Writer. Inputs to the D type flip-flop are connected to DIO0 and DIO1. Hence, choose 'Lines to Write' as 0-3 in the Digital Writer.

Important: You cannot configure one line as both an input and output simultaneously as this will cause a conflict between the Digital Reader and Digital Writer.

- Make DIO 0 and DIO 1 logic low by setting the lines to position "LO" on the Digital Writer. This is done to ensure than the clock and D4 input to the flip-flopare logic low initially.
- Run both Digital Reader and Digital Writer.

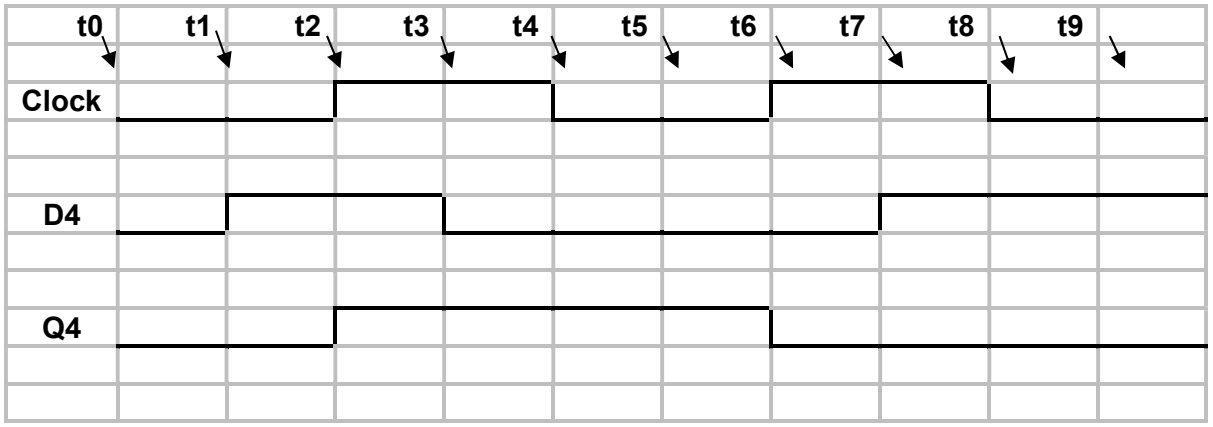
- Carefully record the values on Q4 from the **Digital Reader's line 7** as a result of the change made to line 0 (Clock) and line 1 (D4) on the **Digital Writer** at the **chronological order (time steps)** given in the table below. (Note that LO= logic low and HI= logic high.)

Time step	Inputs (change the values on the Digital Writer as follows)		Output Q4 (read the value from line 7 of Digital Reader)
	D4 (line 1)	Clock (line 0)	
t0	LO	LO	LO
t1	HI	Unchanged from last	LO
t2	Unchanged from last	HI	HI
t3	LO	Unchanged from last	HI
t4	Unchanged from last	LO	HI
t5	Unchanged from last	Unchanged from last	HI
t6	Unchanged from last	HI	LO
t7	HI	Unchanged from last	LO
t8	Unchanged from last	LO	LO
t9	Unchanged from last	Unchanged from last	LO

- Before proceeding with the lab, consider the sequence of transitions that you have just recorded. If these are not what you would expect from a D-type flip flop then you should double check your circuit and procedure of recording.

Task 3
(8 marks)

Using the values you have recorded for the D-type flip-flop, plot the graphs of the inputs (Clock, D4) and output (Q4) logic levels as a function of time.



Task 4
(16 marks)

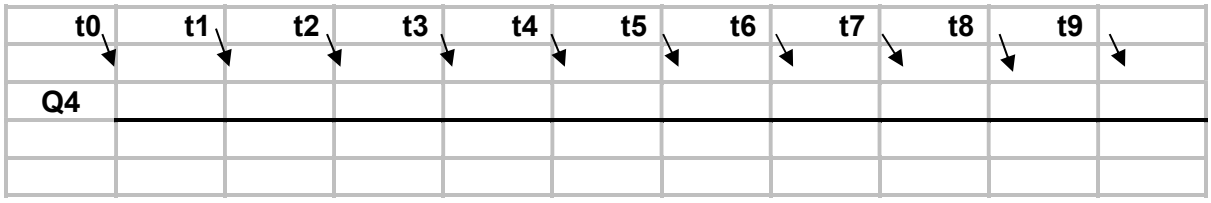
With reference to specific aspects of waveforms that you have plotted in Task 1 and Task 3, explain how the transitions demonstrate the behaviour of a latch and a flip-flop.

Task 1 waveform demonstrates the behaviour of a latch as Q3 updates as D3 changes while clock is high, (output will change multiple times if D changes during high clock phase)

Task 3 waveform demonstrates the behaviour of a flip-flop as Q4 holds its value between rising edges of low to high transition of clock edge

Task5
(6 marks)

Consider the transitions on Clock and D4 for the D-type flip flop you have recorded in Task 3. The D-type flip-flop we have used is positive-edge triggered. What is the output if the flip-flop is negative-edge triggered? Plot the timing diagram for Q4 if the D-type flip-flop is negative edge triggered.



Part C: Sequencer

Task 6

You are to design, build and test your own sequencer. A unique sequencer for you is given below. Use this sequence to complete the tasks given in the table below.

Table: Your assigned unique sequencer

Surname	Sequence of Lab 2
Surname from A to G	S0 S7 S1 S3 S2 S6 S5 S1
Surname from H to Q	S0 S2 S6 S5 S3 S6
Surname from R to Z	S0 S4 S5 S7 S3 S2 S4

You will be using the CD40175BE which is a quad D-type flip-flop integrated circuit. You will also need to use the SN74C00N which is a quad 2-input NAND gate integrated circuit.

You must decide which pins of the chips CD40175BE and SN74C00N are to be used.

You must also decide which connections on the NI myDAQ are to be used to provide the inputs to the chips using the NI ELVISmx Digital Writer and to monitor the outputs from the chip using the NI ELVISmx Digital Reader.

The schematics for the CD40175BE and the SN74C00N are shown below.

Note: It may be necessary to use the inverted outputs of the flip-flops.

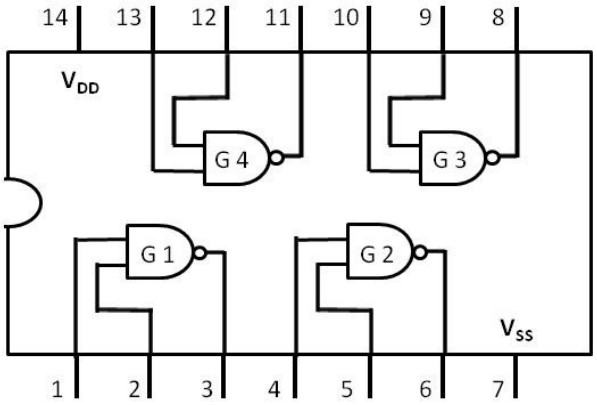


Figure 9: Pin assignment for SN74AC00N

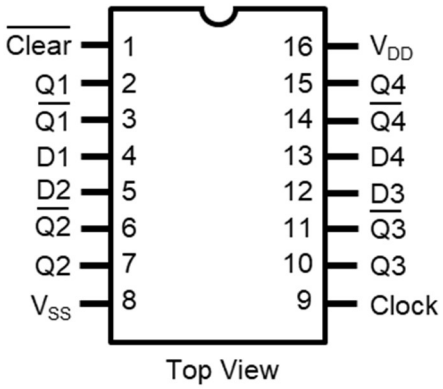
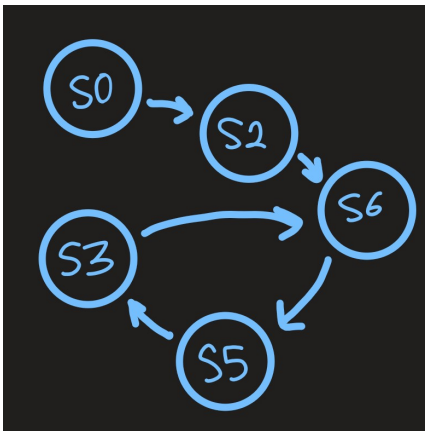


Figure 10: Pin assignment for CD40175BE by Texas Instruments.

Task 6.1
(4 mark)

Draw the state diagram for the sequencer:



Task 6.2
(8 marks)

Complete the state transition table for the sequencer from Task 6.1.

	Present State			Next State		
	Q_3	Q_2	Q_1	D_3	D_2	D_1
S_0	0	0	0	0	1	0
S_1	0	0	1	X	X	X
S_2	0	1	0	1	1	0
S_3	0	1	1	1	1	0
S_4	1	0	0	X	X	X
S_5	1	0	1	0	1	1
S_6	1	1	0	1	0	1
S_7	1	1	1	X	X	X

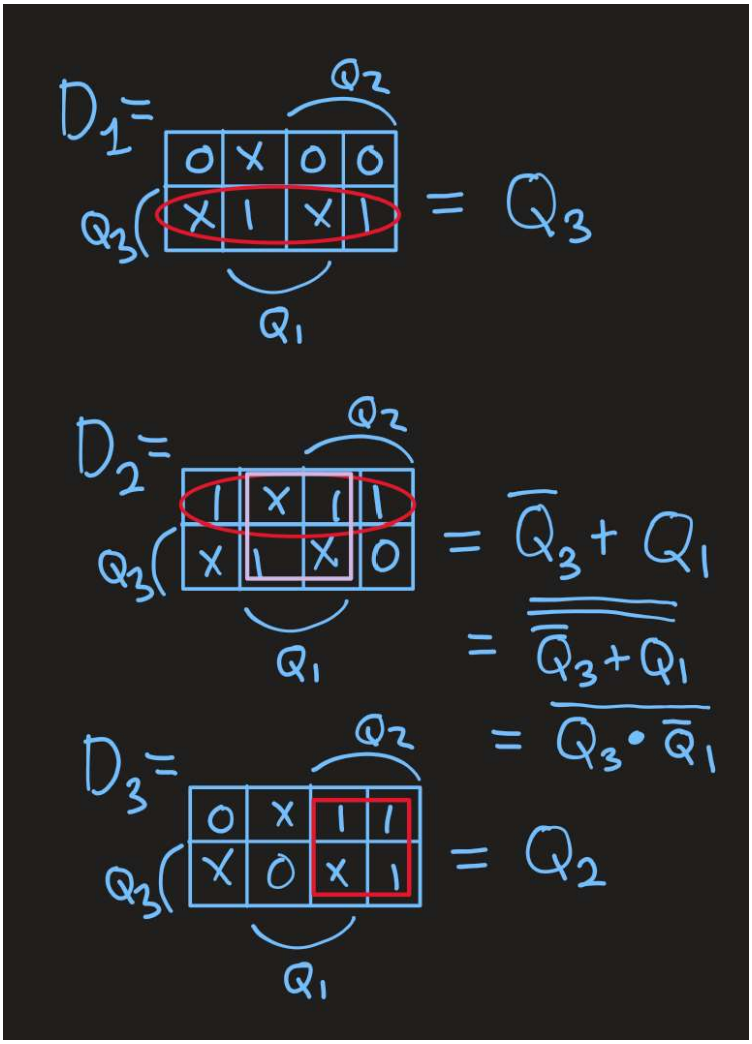
State Transition Table

Task 6.3
(12 marks)

Using Karnaugh maps produce the minimised next state logic for the sequencer. The sum-of-products expressions will have to be converted to NAND-NAND format in order to implement the circuit using the available chips.

Indicate the prime implicants on the Karnaugh maps.

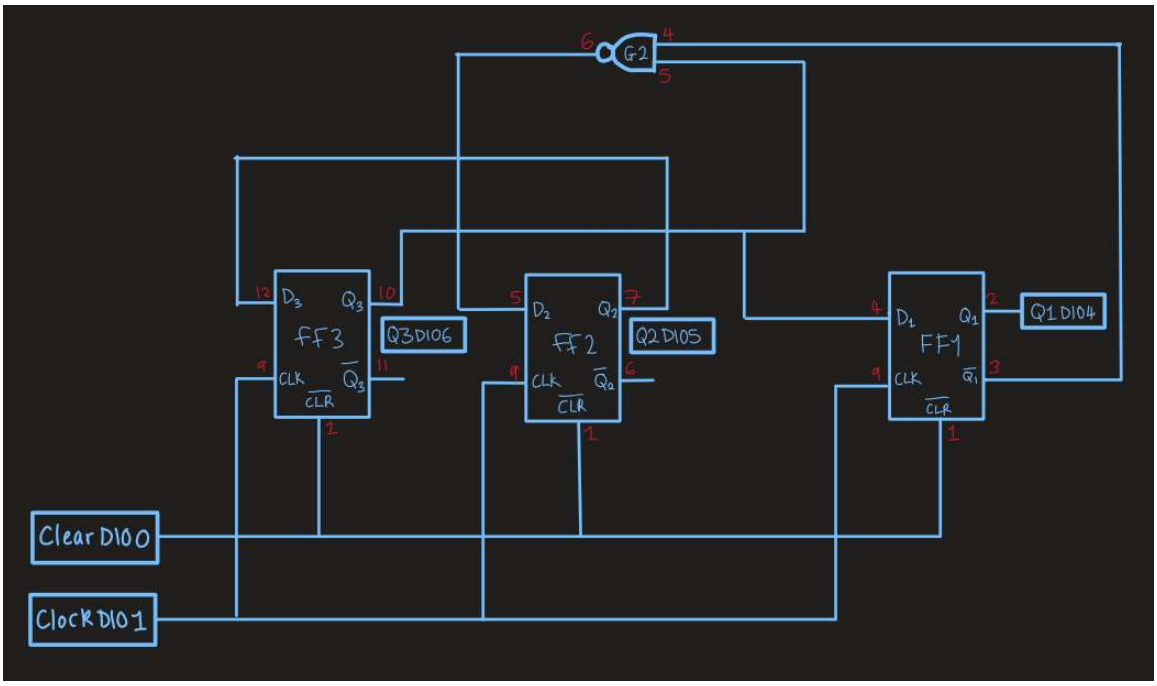
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Task 6.4
(16 marks)

Using the logic equations that you have derived in Task 6.3, produce a circuit diagram for the sequencer using any of the chips that have been used previously in this lab session (you will have to use NAND gates and flip-flops). The diagram should give all the information necessary to enable another person build the circuit on the breadboard of the NI myDAQ.

Refer back to the “Essential Lab Preparation” section to learn which information you should include on the circuit diagram. You will lose marks if this information is not included.



Task 6.5
(16 marks)

Build the circuit form Task 6.4 on your breadboard with connection to MyDaq. Change the clock signal from low to high repeatedly and test whether the sequencer cycles through all the states as given in the state diagram.

Attend the scheduled on-campus lab at allocated one-hour time slot, and demonstrate the circuit (you may assemble the circuit before attending or build the circuit on-site within the time slot). The mark is given at the time of demonstration, which is: 16 /16.

You are only required to attend at your allocated one-hour time slot within your timetable's period.

Surname from A to G: 1st hour of the 3-hour period

Surname from H to Q: 2nd hour of the 3-hour period

Surname from R to Z: 3rd hour of the 3-hour period